

The λ_2 -axis observable sweep — one Schur generator, honestly split into its two legs

Grace | round 6, item 5 | 2026-07-18. The λ_2 (short-root) axis of B_2 is a literal coordinate whose sign is a boundary-support prediction. This sweep enumerates what it sets across sectors — a Schur generator (one property, many observables). Honoring Lyra's F586 retraction: this is ONE engine with TWO legs — the exact leg (boundary vanishing, the clean λ_2 dichotomy) and the graded leg (overlap suppression, a cousin). I keep them separate so the sweep doesn't over-unify.

The property (linear algebra, exact)

An $SO(5)$ K-type is labeled (λ_1, λ_2) , $\lambda_1 \geq \lambda_2 \geq 0$. By the round-4/5 Shilov theorem (Szegő restriction + $SO(5) \downarrow SO(4)$ class-1 branching): a state's Shilov-boundary value is zero iff $\lambda_2 > 0$ (non-spherical); spherical ($\lambda_2=0$) reaches the boundary. The substrate's own $\rho = (3/2, 1/2)$ has $\lambda_2=1/2 > 0$ — the non-spherical direction is built in. λ_2 is the short-root (e_2) axis of the B_2 diagram (Fig 2).

LEG 1 — the EXACT leg (boundary support): the clean λ_2 dichotomy [DERIVED for color]

state	λ_2	Shilov boundary	observable	tier
color singlet (lepton, meson, baryon)	0 (spherical)	reaches it	emitted / free / asymptotic	DERIVED (K745)
color triplet (quark), octet (gluon)	> 0 (non-spherical)	zero support	CONFINEMENT — no free quarks/gluons	DERIVED (3 routes, K745)
photon ($U(1)$, color singlet)	0 (spherical)	reaches it	massless, free, long-range (emitted)	consistent

The exact statement: which states are asymptotic is *exactly* which are color singlets ($\lambda_2=0$). Confinement and emission are the same theorem read on the two sides of the λ_2 divide — one Schur-orthogonality line, frame-independent. This is the derived leg.

❓ CORRECTION (2026-07-18, per Keeper K746 + Lyra): $n(\nu_R)=2$ is NOT a λ_2 result

An earlier version put $n(\nu_R)=2 \rightarrow m_1=0$ under the λ_2 graded leg (" ν_R is spherical $\rightarrow$ skips the $\lambda_2>0$ Shilov stratum"). That premise was RETIRED (Lyra): it conflated the K-type boundary-value λ_2 (the confinement theorem) with the boundary-orbit rank. The correct $n(\nu_R)=2$ uses the idempotent count (the rank-2 Jordan frame's 2 spectral idempotents index the gauge-singlet ν_R) and does not use λ_2 at all. So $m_1=0$ belongs to the domain-rank mechanism, not this λ_2 sweep. Removed from Leg 2 below.

LEG 2 — the GRADED leg (overlap suppression): a COUSIN, not the same theorem [structural]

These trace to bulk-overlap suppression *along* the Wolf strata — correlated with λ_2 but a distinct mechanism (Lyra F586: do NOT conflate with Leg 1). ($n(\nu_R)=2/m_1=0$ removed — it's the domain-rank/idempotent mechanism, not λ_2 .) | state / quantity | stratum / λ_2 role | observable | tier | |—|—|—|—| gen-1 (bulk / origin) | deep bulk, min boundary reach | electron-at-origin; V_{us} pure-radial; m_1 lightest | structural | | gen-3 (Shilov stratum) | $\lambda_2>0$, boundary edge | τ (heaviest lepton) | structural | | 2nd-neighbor overlap (bulk $\leftrightarrow$ Shilov) | crosses the λ_2 divide | V_{ub} smallest (softest CKM element) | structural |

The one corollary that spans both legs (already rendered)

The CKM-small / PMNS-large asymmetry is which strata a sector populates: quarks fill all 3 boundary orbits (Dirac-aligned $\rightarrow$ small CKM); neutrinos are indexed by only the 2 spectral idempotents (gauge singlets, $n(\nu_R)=2$ — the domain-rank/idempotent count, NOT λ_2) $\rightarrow$ different structure $\rightarrow$ large PMNS. One geometric fact (the 3-orbit vs 2-idempotent split), rendered by one overlap-matrix machine (CKM F585 + PMNS F564/F584).

Net (Schur-generator, honestly tiered)

The single property “ λ_2 spherical vs non-spherical” is a Schur generator — one coordinate axis setting many observables. But it is honestly two legs: - Exact leg (λ_2 boundary dichotomy): confinement + emission/free states + the photon — DERIVED (color). - Graded leg (strata overlap suppression): V_{ub} softness, the gen-1/gen-3 mass ordering — structural, a *cousin* of the exact leg. ($m_1=0 / n(\nu_R)=2$ is NOT here — it’s the domain-rank/idempotent mechanism, see correction above.)

λ_2 is a physical prediction for every state — its sign says “confined or free.” That is the compression: the exact leg of the Standard Model’s confinement is a single coordinate on the B_2 root diagram. (Pairs with the two other master mechanisms — odd-g and the domain-rank massless-mode count, item 3.)

— Grace, 2026-07-18. λ_2 -axis sweep: Leg 1 (boundary support) = confinement/emission DERIVED; Leg 2 (overlap) = mass-sector cousin, structural/supported. Kept the two legs separate per F586. λ_2 = the B_2 short-root axis = a boundary-support prediction per state.